

MTN GHANA TEKYEREMA PA HACKATHON 2026

Tie Me Ghana

Hospital communication for Deaf and Hard of Hearing patients. How it is built, how it works, and how the two devices talk to each other.

THE PROBLEM

The literacy trap

A Deaf patient cannot hear their name called, cannot explain what happened after an accident, and cannot count on an interpreter for a walk-in visit.

Pen and paper is not the fallback

Fluency in Ghanaian Sign Language does not imply fluency in written English. Writing a question down excludes exactly the people the system exists to serve.

The problem was identified by the Deaf member of our own team, from personal experience. The published research corroborates it rather than originates it.

What that forces on the design

- The first question cannot be asked in writing.
- Every answer must reach the clinician **as sound**, because their eyes are on the patient.
- A missing word can change a dose. Silence is safer than a guess.
- It has to work on a hospital connection, and afterwards with none.

WHAT IT DOES

Two directions, and one branch

Doctor → patient

Typed or spoken English or Twi is captioned, then tokenised and matched against reviewed GhSL clips, played as one sign video. A word with no sign is fingerspelled.

Patient → doctor

Every reply, typed or tapped, is **spoken aloud** in Twi or English on the doctor's device. The doctor hears the answer without looking at the screen.

The branch

A sign-video question asks whether the patient reads. Literate: free captioning and typed replies. Otherwise: Guided Interrogation, a reviewed question bank answered by tapping.

Emergency Visual Triage

Reachable before any question is answered. Pain scale, body map, critical alerts. Needs no footage and no literacy answer, because a patient who cannot breathe is not asked whether they read first.

The prescription

Medicines become sign video plus a photograph of the box, taken home by QR code or saved to the phone as a file that plays with no network.

PART ONE

The architecture

Four pieces, and one deliberate absence.

What runs where

The app · Netlify

React 19 and Vite 7, shipped as a PWA. Installable, and it keeps the sign videos on the device so a prescription replays at home.

The API · Render

Django 5.2 and DRF. Five apps: `clips`, `consultations`, `prescriptions`, `pairing`, `core`.

Data · Neon + R2

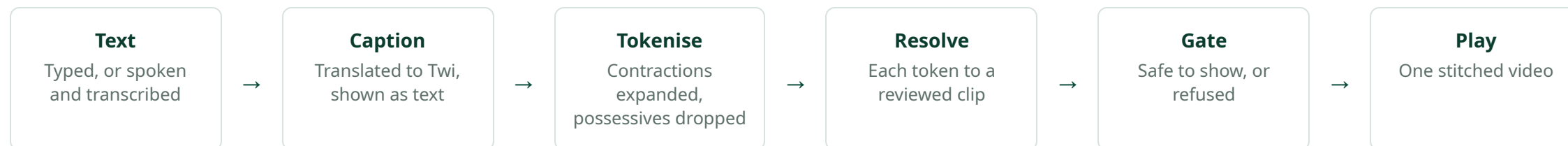
Postgres on Neon for the clip library and prescriptions. Cloudflare R2 for the video itself, because a container filesystem is lost on every restart.

The absence

No consultation content is ever stored on the server. Captions, answers and transcripts live on the devices. The API holds the clip library, issues prescriptions, and brokers a pairing handshake. It has no endpoint that could accept a transcript, and a test asserts that.

There is no login. Nothing here reads `request.user`, and an account would defeat a patient replaying their own prescription months later on their own phone.

From a sentence to a sign video



Resolution order

- **Phrase** — a whole sentence filmed as one clip wins over stitching its words.
- **Word** — the gloss itself.
- **Reviewed alias** — a consultant's judgment that two words are one sign.
- **Plural reading** — "brings" reaches BRING, because GhSL does not mark plural.
- **Fingerspelling** — letter by letter.

Two queries, whatever the sentence

Lookups are batched: word glosses and their readings in one query, aliases only for what missed, the alphabet only if something needs spelling, and the phrase clips. A test pins the count so a long sentence cannot become a query per word.

A missing word can change the dose

A word that cannot be signed used to be simply absent, and absence changes meaning. "Do you have no pain" becomes "pain". "Take two tablets" becomes "tablets".

Droppable

Articles and copulas. GhSL does not use them, so leaving one out is more natural, not broken. Reported to the doctor anyway.

Blocking

Negations, quantities, frequencies, severities. If one cannot be signed **the sentence is refused** rather than shown without it. Never fingerspelled.

Content

Carries the meaning. Fingerspelled if the alphabet is filmed; otherwise the sentence is reported as incomplete.

The doctor reads back what the patient will actually see — the glosses, not the typed words — before anything is shown. An unreviewed clip can never reach a patient: the resolver only ever sees clips a GhSL-fluent consultant has approved and that have footage.

Filmed once, reviewed, made small

70

clips ready for clinical use

102 MB

of footage as filmed

8.5 MB

after compression on upload

4

kinds: word, phrase, letter, alert

Every clip is gated twice

- It must be **filmed** — a row can exist before its footage does, so the library doubles as the list of what still needs recording.
- It must be **approved** by a named GhSL consultant. New footage resets the approval, because the approval was for the recording, not the word.

Replacing the library is one command

`replace_clip_library` deletes the old footage from the database and the bucket, imports a folder, compresses on the way in, and guesses word, phrase or letter from the filename. It prints what it would change and stops unless told `--yes`.

Two files with identical bytes are imported but left awaiting review: two different signs are usually not one recording.

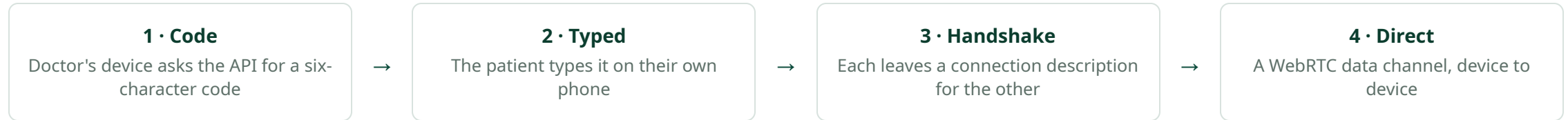
PART TWO

How the two devices talk

A visit can run on the doctor's device and the patient's own phone.

PAIRING

The server introduces them, then gets out of the way



What the server sees

- A six-character code, ten minutes, one pairing.
- Two connection descriptions: how to reach a device, not anything said.
- A rendezvous token for reconnecting, at most four hours.

All of it in a process cache with a timeout. No models, no migrations, nothing written to disk.

No relay, and an honest failure

A TURN server would route the consultation through a third party, which is the thing this design refuses. Some networks will therefore not connect. The app says so and offers to share one device, rather than quietly sending the consultation another way.

What actually crosses the channel

MESSAGE	DIRECTION	WHAT IT CARRIES
literacy	doctor → phone	Ask the literacy question here too
literacy-answer	phone → doctor	Yes or no; the doctor's device records it
path	doctor → phone	Which consultation this is: literate or guided
question	doctor → phone	A caption that has already cleared the safety gate
answer / reply	phone → doctor	A tap or typed reply, to be spoken on the doctor's device
answered	doctor → phone	This question is closed; whichever device answered first won
speaking	doctor → phone	working / playing / spoken / stopped / blocked / failed
emergency	doctor → phone	Follow to the emergency screen, and back
triage	phone → doctor	Which button was tapped — never the words
ended	doctor → phone	The visit is over

The channel hands a screen only its newest message, so every message that says *where the phone should be* also carries the path, the way back, and whether emergency mode is open. A message that collapses into another cannot leave the phone on the wrong screen.

The split is clinical, not technical

On the doctor's device

- The safety gate. Nothing reaches the phone until it has cleared.
- **The sound.** A reply is spoken where the doctor is standing; audio never crosses the channel, only the text does.
- Recording what the doctor personally observed — a nod, a shake of the head.

On the patient's phone

- The literacy question, with the sign video and both answer signs.
- Yes and No, where it hurts, the emergency taps, a typed reply.
- Its own transcript, which stays on that phone until its owner deletes it.

Either device may answer a yes-or-no question. **Whichever comes first is the answer**, and the record says who gave it, so a transcript can never imply the patient tapped something they never touched.

A reload is not the end of a visit

Coming back

Once connected, the doctor's device mints a rendezvous token, registers it, and hands it over. Both keep it. A refresh on either side reconnects by itself under that token, with no new code.

MEASURED 1.6 s for the patient, 1.6 s for the doctor, 1.7 s for a first pairing.

What made it fast

- Post the connection description as soon as a usable address has arrived and gone quiet, rather than waiting out the whole gathering budget.
- Look for the other device every 300 ms at first, not every 1.5 s.
- Treat "not found" on a token as *not yet*, not as failure.

The doctor's screen never moves while the phone is away. It says the phone is away, and the phone rejoins on its own. Only the doctor starting a new patient ends the visit.

The connection is the scarce thing

11 ms

a clip already on the device

~0.5 s

a clip fetched now

~150 KB/s

the line we measured

8.5 MB

the whole library

Warming the library

On first open, every clip is pulled onto the device by a service worker, highest value first: prompts, then emergency alerts, then the alphabet, then everything else.

The bug that taught us the rule

The warm-up was taking the whole connection from the clip a patient was waiting for — over twenty seconds for one sign. It now **stands aside** while any video is loading. An optimisation must never be the reason someone is kept waiting.

The prescription is the payoff: the patient's phone keeps the videos, and saves them as a file that plays months later with no network at all.

PART THREE

How it was built

And how we keep it honest.

THE METHOD

Decisions are written down, and verified in real browsers

61

architecture decision records

654

backend tests

1,278

frontend tests

90

commits on the branch

Every non-obvious choice has an ADR

What was decided, what it cost, and what it rules out. Including the mistakes: a clip overwritten in the bucket, a test that passed locally and failed in CI, a cache that leaked between tests.

Tested where it actually runs

Two real headless browsers over real WebRTC for pairing; the real bucket for offline replay; timings measured rather than estimated. A green unit suite is not evidence that two devices connect.

CI runs lint, both test suites, a migration check and a build on every push.

What is not done

The alphabet is not filmed

Thirty-six letter clips. Until they exist, fingerspelling cannot complete, so any word with no sign still refuses the sentence. It is the single highest-value footage left, because twenty-six clips cover every medicine name that will ever be prescribed.

The body-location grid

Withheld until all sixteen are filmed: a patient offered three body parts when the pain is in a fourth taps the nearest one, and that wrong answer looks exactly like a right one.

Needs a clinician's review

- The safety word lists are a starting point, not a clinical judgment.
- Every sign in the library, and the two pairs that share one recording.
- The emergency alerts, before any public demo.

Known limits

- No TURN, so some networks will not pair.
- Anyone who types the code while it is showing joins the visit.
- The pairing cache is per process: the API must run as one.

IN ONE SENTENCE

The server never learns what was said.

Everything else — the clip library, the safety gate, the direct connection, the offline copy — exists to make that possible without making the consultation worse.